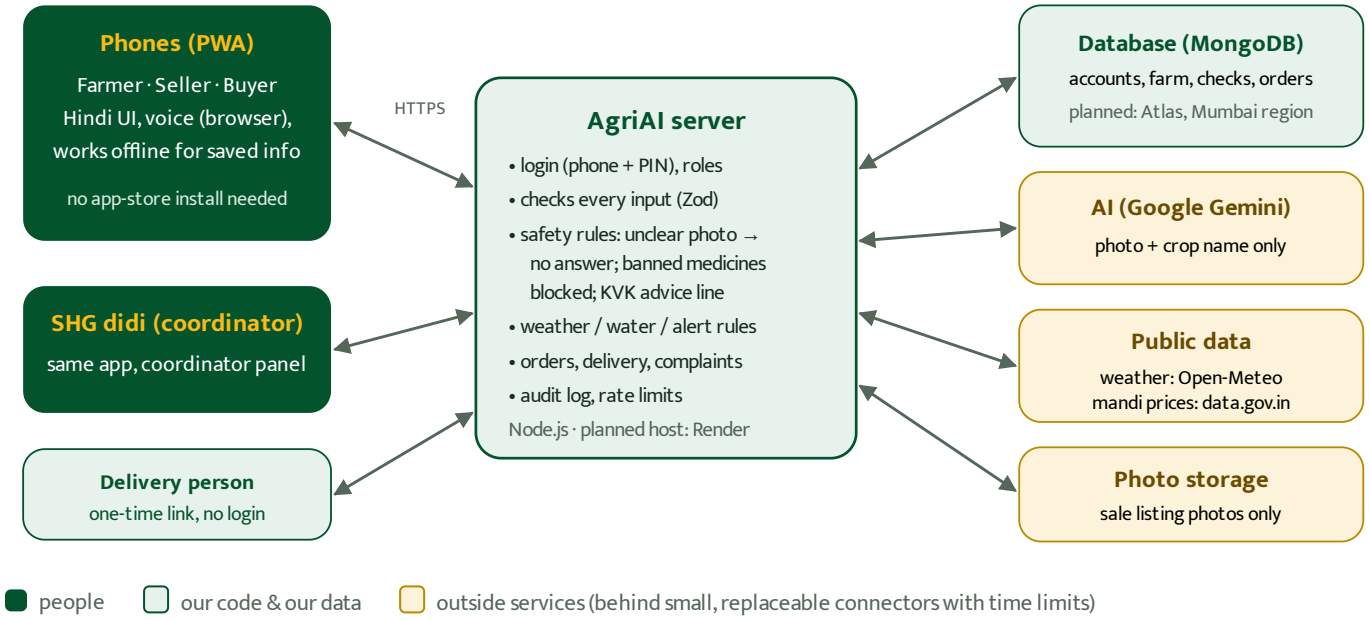
One Hindi web app on the farmer's own phone, one small server, one database. AI is used only through one narrow, replaceable connection – and it never receives a farmer's name, phone number or location.



In plain words

1. **The farmer opens a link** in Chrome and can add it to the home screen. Everything is in simple Hindi, with 🔊 read-aloud and a mic button (browser speech).
2. **Photo check:** the phone shrinks the photo (and removes GPS data), the server sends only the photo and the crop name to the AI, then applies our safety rules before showing anything.
3. **Weather, water, calendar, alerts** are rules written down in one place (marked "verify with KVK"), not AI guesses.
4. **Marketplace:** listings, orders, delivery tracking, complaints and the commission ledger run on our server. Money moves directly between buyer and seller (UPI or cash).
5. **SHG didi** uses the same app with a coordinator panel; every coordinator action is written to an append-only audit log.

Design choices that matter for government

- **Replaceable AI:** all AI calls go through one interface. Another model (including an Indian or government-hosted one) can be connected without changing the app.
- **Low cost, low bandwidth:** a web app (no app store), small first load, offline screen for saved advice.
- **Works without the AI:** if the AI is slow or down, weather, water, market and coordinator work continue.
- **Offline demo mode:** recorded real AI answers for demonstrations; it cannot be turned on in production.
- **Open, testable code:** 566 automatic server tests, 146 app tests and 8 end-to-end phone journeys run before each change.

Status: built and tested with demo data on a laptop. Cloud deployment (Render, Atlas) is prepared but not yet live; no real users yet.